

Bluestock Fintech Capstone

Mutual Fund Analytics Platform

Final Project Report

Datasets: 10 CSV files | 40 Schemes | 10 AMCs

Tech: Python Pandas SQLite Plotly scikit-learn

1. Introduction

This report presents the Bluestock Fintech Mutual Fund Analytics Capstone. The platform delivers end-to-end analysis of the Indian mutual fund industry covering ETL processing, SQL analytics, EDA, risk-return metrics, and a fund recommendation engine.

2. Objective

1. Build a production ETL pipeline for 10 mutual fund datasets.
2. Compute CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown.
3. Rank 40 schemes with a weighted composite scoring model.
4. Develop a risk-profile recommendation engine (Low/Medium/High).
5. Deliver reproducible notebooks, scripts, and SQL queries.

3. Datasets

File	Rows	Description
01_fund_master.csv	40	Scheme metadata
02_nav_history.csv	46,000	Daily NAV 2022-2026
03_aum_by_fund_house.csv	90	Bi-annual AUM per AMC
04_monthly_sip_inflows.csv	48	Industry SIP stats
07_scheme_performance.csv	40	Risk-return metrics
08_investor_transactions.csv	32,778	Investor transactions
10_benchmark_indices.csv	8,050	NIFTY50 + 6 indices

4. Data Cleaning

Cleaning via scripts/etl_pipeline.py and notebooks/02_data_cleaning.ipynb.

NAV: dates parsed, daily_return_pct and rolling_vol_30d computed.

Performance: risk_score (1-6), excess_return, information_ratio derived.

Transactions: date parts, amount_bucket, kyc_verified flag.

Quality: 9/10 datasets zero nulls. 1 WARN (yoy_growth_pct - expected).

5. Database Design (bluestock_mf.db)

Table	Description
dim_fund	Scheme attributes (amfi_code PK, n
fact_nav	46,000 rows daily NAV
fact_performance	40 rows risk-return metrics

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fact_transaction	32,778 investor transactions
fact_aum	90 rows AUM by fund house

6. Exploratory Data Analysis

- * SBI MF leads AUM at Rs 12.5 lakh crore (Mar 2025).
- * SIP inflows peaked at Rs 31,002 Cr in Dec 2025.
- * Folio count doubled: 13.26 Cr to 26.12 Cr (2022-2025).
- * BFSI + IT dominate equity portfolio holdings.
- * 26-35 age group: highest transaction count (13,463).
- * T30 cities: 66% of all investment transactions.
- * 20+ charts generated across EDA notebook.

7. Performance Analytics

Metric	Mean	Best Scheme
3Y CAGR	14.09%	SBI Small Cap Regular (23.39%)
Sharpe Ratio	0.54	Mirae Asset Large Cap (1.45)
Sortino	0.92	Mirae Asset Large Cap (2.39)
Max Drawdown	-17.87%	ICICI Pru Liquid (-0.10%)
Alpha	1.25	HDFC Short Term Debt (1.98)

8. Advanced Analytics & Fund Scoring

Composite score = $0.30 \times \text{return_rank} + 0.25 \times \text{sharpe_rank} + 0.20 \times \text{alpha_rank} + 0.15 \times \text{expense_rank} + 0.10 \times \text{dd_rank}$

Scores normalised 0-100 (MinMaxScaler). 100 = Best fund.

Rank	Scheme	3Y Ret	Sharpe	Score
1	SBI Small Cap Fund - Direct Plan -	23.14%	0.930	100.0
2	Kotak Flexicap Fund - Regular - Gro	15.65%	0.980	97.4
3	Kotak Emerging Equity Fund - Regula	18.23%	0.960	95.4
4	ABSL Small Cap Fund - Regular - Gro	22.38%	0.900	90.7
5	SBI Small Cap Fund - Regular Plan -	23.39%	0.940	82.7

9. Recommendation Engine

recommender.py -> recommend('Low' | 'Medium' | 'High')

Low Risk (risk_score <= 2): HDFC Short Term Debt score=79.9

Medium (risk_score <= 4): Kotak Flexicap score=97.4

High (all funds): SBI Small Cap Direct score=100.0

Ranking: fund_score_100 -> Sharpe ratio -> 3-year return.

10. Conclusion

- * 40 schemes profiled across 10 AMCs with full risk-return metrics
- * ETL pipeline: 87,533 rows processed in 1.8 seconds
- * 5 notebooks: ingestion->cleaning->EDA->performance->advanced

- * 8 reusable financial metric functions (compute_metrics.py)
- * SQL schema + 10 business queries
- * Top 5 funds: +112% vs NIFTY50 +5.9% over same period
- * Final validation: 38/38 checks passed -- PROJECT COMPLETE